



Riddhi's Birthday World: Codex-Ready Build Spec (v2)

How to use with Codex: paste Sections 1 to 4 once at the start of the session as standing rules. Then give it **one Stage prompt at a time** (Section 12). Never ask for everything at once. After each stage, test on a real phone and only then continue.

Product in one line: a mobile-first static website, unlocked at **2026-10-15 00:00 IST**, that guides Riddhi (turning 22) through a night-sky birthday journey led by a dog guide. Dark navy + aurora theme. No backend, no framework.

1. Tech stack (fixed, do not change)

- Plain **HTML + CSS + vanilla JS (ES modules)**. No React, no build step, no npm required.
- **GSAP 3.12** (core only) from cdnjs for sequences. **canvas-confetti 1.9** for bursts. Both from `cdnjs.cloudflare.com`, pinned versions.
- Fonts: **Fredoka** (headings, playful) + **Nunito** (body) from Google Fonts, with fallback `system-ui, sans-serif`. Preload them before showing UI.
- Hosting: Netlify drop. Single-page app: one `index.html`, chapters are `<section>` elements toggled by JS.
- Wish delivery: **Web3Forms** (free) via `fetch` POST. Key goes in `config.js`.
- Target: iPhone Safari 15+, Android Chrome 100+, desktop Chrome. Portrait first. Total weight under 25 MB.

2. Folder structure (exact)

```

riddhi-birthday/
├── index.html
├── css/ tokens.css base.css chapters.css animations.css
├── js/  config.js main.js state.js preloader.js audio.js timelock.js
      |  aurora.js stars.js confetti.js dog.js mic.js wish.js
      |  chapters/ c00-lock.js c01-loader.js c02-entrance.js c03-identity.js
      |              c04-cake.js c05-wish.js c06-balloons.js c07-cards.js
      |              c08-memories.js c09-games.js c10-vouchers.js c11-eggs.js
      |              c12-voice.js c13-letter.js c14-finale.js
├── audio/ birthday-tune.mp3 pop.mp3 whoosh.mp3 sparkle.mp3 blow.mp3 match.mp3 whistle.mp3 sn
├── dogs/  dog-01.png ... dog-10.png (transparent PNG, 300px wide; use simple placeholders u
└── others/ photos/ (01-same-bus.webp ... 10-gifts.webp + extra-01..06.webp)  voice/ (voice-no

```

Missing files must never crash the app: use a placeholder (soft gradient + dog emoji) and `console.warn`.

3. Global rules (non-negotiable)

1. Animate **only** transform and opacity (plus `filter: blur` only on the static aurora layers). Never animate `top/left/width/height`.
2. Add `will-change: transform, opacity` only during an animation, remove after.
3. **Preload everything** before Chapter 1 ends (Section 6). No image may appear half-loaded: show elements only after `img.onload()`.

elements only after `img.decode()`.

4. Particle budget: stars 80 (mobile) / 160 (desktop); confetti `particleCount` max 120 mobile; use `disableForReducedMotion: false`.
5. One `requestAnimationFrame` loop for background canvases; pause it when the tab is hidden (`visibilitychange`).
6. Chapter transition: outgoing fades + drifts up 20px (0.5s, `power2.in`), incoming fades from 20px below (0.7s, `power3.out`). Never snap.
7. Easings: entrances `power3.out`; buttons/pops `back.out(1.7)`; flips `power2.inOut`; floaty things `sine.inOut` yoyo.
8. All tap targets at least 48x48px. Use `touch-action: manipulation, -webkit-tap-highlight-color: transparent, user-select: none, 100dvh (not 100vh), env(safe-area-inset-*)`.
9. No text smaller than 16px. No horizontal scroll (`overflow-x: hidden` on body).
10. Wrap each chapter's `init()` in try/catch. On error, log it and show a "Skip →" button so the story never dead-ends.
11. **All copy lives in** `config.js`. No user-facing string is hardcoded in chapter files. The owner edits only `config.js` and others/.
12. Every chapter exports `{ id, init(ctx), destroy() }`. `destroy()` kills GSAP timelines, event listeners, intervals.

4. Design tokens (`tokens.css`)

```
:root{
  --bg-0:#050b1f; --bg-1:#0a1840; --bg-2:#10265f;
  --blue:#4da3ff; --sky:#9fd0ff; --ice:#e8f3ff;
  --aurora-g:#3dffb0; --aurora-t:#2bd6e8; --aurora-v:#8f6bff;
  --gold:#ffd36b; --pink:#ff7eb6; --danger:#ff6b6b;
  --card:rgba(255,255,255,.08); --card-border:rgba(255,255,255,.18);
  --radius:22px; --glow:0 0 24px rgba(77,163,255,.55);
  --font-h:'Fredoka',system-ui,sans-serif; --font-b:'Nunito',system-ui,sans-serif;
}
```

Cards: glassmorphism (`backdrop-filter: blur(12px), --card, 1px --card-border`). Primary buttons: gradient `--blue` to `--aurora-v`, rounded pill, soft glow, press = `scale(.95)`. **Dog sticker rule:** every card has a dog `<img class="dog-top">` positioned `top:-38px; left:50%; transform:translateX(-50%), 76px` wide, overlapping the card edge, with a 1.2s gentle wobble (`rotate(-4deg)` to `4deg`, yoyo). Cycle through dog-01..10 so neighbours differ.

5. App core

`state.js`: { `unlocked, audioOn, muted, chapter, stamps:Set, eggsFound:Set, wishSent` }. Persist `stamps, eggsFound, wishSent` in `localStorage` (try/catch), so a reload resumes gracefully. `main.js`: boots, runs `timelock`, starts `preloader`, then chapter controller `go(id)`. Fixed layers (z-index order): `#aurora` canvas (0), `#stars` canvas (1), `#stage` chapters (10), `#hud` (50: mute button top-right, paw-stamp bar top-center, replay menu bottom, shown from Chapter 3 onward), `#toast` (60). **Paw stamps:** awarded at end of chapters 3,4,5,7,8,9,10,11 (8 total). Bar shows 8 slots; earned slot pops with `back.out` + sparkle. All 8 = "Gold Level".

6. Preloader (`preloader.js`)

- Manifest built from `config.assets` (images, audio, dogs, fonts). Load images with `new Image()` +

```
await img.decode(); audio with fetch then blob (or canplaythrough); fonts with
document.fonts.load.
```

- Emit progress 0..1 (weighted by count). Chapter 1 heart fill maps to this. Minimum display time 2.5s so the splash is visible.
- Timeout per asset 15s: on failure mark as placeholder, continue.
- Chapter 0 (lock) must work **without** waiting for the full preload. Start preloading silently there, so by midnight everything is ready.

7. Time lock (timelock.js)

```
export const TARGET = new Date('2026-10-15T00:00:00+05:30').getTime();
```

- Countdown ticks every 250ms using `Date.now()`; display D:H:M:S. When `now >= TARGET`: fire `unlock()`.
- Support `?preview=1` in the URL to bypass the lock for the owner's testing. **Remove or change this before the final deploy.**
- Honest limitation: a client-side lock can be bypassed (devtools). Final protection = deploy the real site only on 14 Oct night; before that the live URL serves a stripped `index.html` containing only Chapter 0.

8. Audio engine (audio.js)

- One `HTMLAudioElement` for the birthday tune: `loop = true`, `preload = 'auto'`, volume 0.55. **Never stops or restarts between chapters.**
- Browsers block autoplay: the first user tap (the "Tap to begin ❤️" button in Chapter 0 / after unlock) calls `audio.play()`. If it was already started on the lock screen, keep playing; at unlock the tune is simply already running. If she arrives after midnight, the same button starts it.
- **Ducking:** `duck()` tweens volume to 0.15 over 0.4s when voice-note or secret voice plays; `unduck()` returns to 0.55 on ended.
- SFX: small pool of preloaded `Audio` clones (`play()` resets `currentTime`). Mute button toggles `muted` for tune + SFX. Mute persists in state only.
- Handle iOS: create/resume an `AudioContext` on first tap if used for mic.

9. Visual engines

Aurora (`aurora.js`): canvas, 3 ribbons (green, teal, violet). Each ribbon: a vertical-gradient band whose centre-line is a sum of 2 sines with different speeds, drawn at half resolution and CSS-scaled up, `filter: blur(40px)`, `globalCompositeOperation: 'lighter'`, alpha 0.25 to 0.4. Slow drift (about 0.15 rad/s). Over a `--bg-0 → --bg-2` linear gradient. 30fps cap on phones. **Stars** (`stars.js`): canvas, 80/160 stars, random radius 0.5 to 1.8, twinkle by sine alpha, 1 shooting star every 6 to 12s (a line with fading tail, 700ms). Exposed `shootingStar.onTap()` for Easter egg 2. **Dog** (`dog.js`): helper `speak(text, {mood})` renders a speech bubble (typewriter 25ms/char, auto-dismiss or tap) beside a dog image. `mood`: idle (bob), jump, sneeze, party-hat. All via transforms.

10. Chapter specs

Each lists **Behaviour** → **Animation** → **Acceptance test**.

C0 Lock. Aurora + stars, sleeping dog (breathing scale 1 to 1.03, "Zzz" glyphs floating up). Title

"Something special is waking up at 12:00 AM...". Countdown below. Dog taps: 1 to 2 snore sfx; 3rd shows `lock.tap3`; 6th shows `lock.tap6`. Button "Tap to begin 🧡" (pulsing glow) starts music. At unlock the dog stretches (scaleY 1.1, rotate) and the scene fades to C1. ✅ Countdown reaches 0 and auto-advances without refresh.

C1 Loader. SVG heart outline; a `clipPath` rect translating upward = fill %. Message rotates every 1.6s with crossfade. At 100%: 2 heartbeats (scale 1.15 , 0.3s), a ring expands and fades, cross-dissolve to C2. ✅ No image visible before loading completes.

C2 Entrance. Giant countdown 10 to 1 (each number: scale 1.6 to 1, fade, tiny pop sfx; whistle sfx starts at 3). Fireworks: `canvas-confetti` shapes: ['circle'], bursts at random points x 0.15 to 0.85, y 0.2 to 0.5, 6 bursts over 2.5s. "HAPPY BIRTHDAY RIDDHI": split into spans, stagger 0.06s, each drops from -80px with `back.out`, gold glow. Confetti cannons from both bottom corners. Golden-retriever dog (party hat) jumps in with `entrance.dog` bubble. Gift-box button "Open your surprise 🎁" (shake every 3s: rotate $\pm 6\text{deg}$, 0.5s). ✅ Music already playing and not restarted.

C3 Identity. Three screens in sequence. Q1/Q2/Q3 from `config.identity`. For the "No" button on Q2: on `touchstart/pointerenter` within 60px it moves to a random position within the card (GSAP `x,y` 0.25s `power2.out`); caught after 5 attempts shows "Nice try 😊". After Q3: scan animation (a laser line translates top to bottom 1.4s over her name card, 2 passes), then **stamp** ("VERIFIED") scales from 3 to 1 with rotate -12deg, plus thud sfx and screen shake ($\pm 4\text{px}$, 0.2s). Award stamp #1.

C4 Cake. Steps are separate taps; each step button hidden until previous completes.

1. Cake (SVG layers: plate, 2 tiers) drops from -300px, squash (scaleY .85 then 1).
2. Decoration: frosting swirl stroke-dashoffset draw 0.8s; 24 sprinkles fall (stagger); 5 cherries pop; banner unfurls (scaleX 0 to 1, origin left).
3. Candles: 22 small candles stagger up (0.04s) OR two number candles "2"+"2" (default: number candles).
4. Light: match sprite flies along arc (bezier via GSAP `motionPath`, or x/y keyframes), flame = animated SVG teardrop (scale + skew flicker, random 80 to 140ms), warm radial glow behind cake (opacity flicker), stage brightens.
5. Countdown 3,2,1 then "BLOW!". Mic path in `mic.js`: `getUserMedia({audio:true});` `AnalyserNode`, `fftSize 256`; blow = RMS above 0.18 for at least 250ms. Always show a "Tap to blow 🌬️" fallback button. If permission denied, silently use tap only.
6. First blow: flames shrink and extinguish, smoke wisps rise (3 translucent blobs); after 1.2s flames **re-ignite** (pop), dog bubble `cake.party`. Second blow extinguishes permanently.
7. A slice slides out of the cake (`translateX`), then a cartoon fork enters, steals it (slice and fork exit right), text `cake.slice` appears.
8. Explosion: 3 confetti bursts, 20 streamers (long thin rects falling with sway), 10 balloons rising off-screen. Stamp #2. ✅ Works with mic denied; works if she blows twice quickly.

C5 Wish. Aurora at max intensity (alpha +0.1). Glass card: `textarea` (max 280 chars), button "Send to the stars 🌟". On send: validate non-empty; text shrinks into a glowing orb (text fades, orb scale 0 to 1, glow), orb rises along a curved path up and out while leaving sparkle trail, then becomes a permanent small star in the sky canvas. Disable the button. `wish.js` posts to Web3Forms: { `access_key`, `subject`: "🌟 Riddhi's birthday wish", `message`: text, `from_name`: "Birthday Site" }. On failure: save in `localStorage.pendingWish` and retry on next load/online event; **UI never shows an error**. Note the disclosure line `wish.note` is shown under the box. Stamp #3.

C6 Balloons. 10 balloons, SVG, colours cycled, absolute-positioned across a 100%-width lane, each floats up/down (`sine.inOut yoyo, 3 to 5s`, different delays) with sway rotate $\pm 3\text{deg}$. Tap: balloon scales 1.15 then bursts (shape vanishes, 12 confetti particles in its colour at its position, pop sfx), message card scales in from 0.6 (`back.out`) in centre with the dog; tap card to dismiss. After 10 popped: `balloons.done`. Message 10 triggers egg hint. Prevent double-tap bugs: set popped flag immediately.

C7 Flip cards. Horizontal scroll-snap carousel (or swipe) of 11 cards, 3D flip (`transform: rotateY(180deg), backface-visibility:hidden, parent perspective:1000px`), 0.7s. Front = nickname (big, gradient text) + emoji; dog on top. Last card flips twice (second flip tap shows `cards[10].back2`). Dots indicator. Stamp #4 once all cards flipped at least once.

C8 Memories. Polaroid stack: 10 cards stacked with slight random rotation ($\pm 6\text{deg}$). Swipe: pointer drag; release beyond 90px or velocity > 0.5 flings card off ($x \pm 600$, rotate $\pm 25\text{deg}$, 0.5s) and next scales up. Card: photo (object-fit cover, 4:5), tape strip, tag sticker bottom (`tag`), caption (`caption`), dog on top. Tap photo = lightbox zoom (scale from card rect, FLIP technique). After last: `memories.end`. Stamp #5.

C9 Games. Menu of 4 tiles; each game has Skip. Stamp #6 after any one game completed or after Skip-all.

- *Match*: 12 cards (6 photo pairs, from `extra-*` photos), flip 3D, mismatch flips back after 700ms, lock input during compare, win = confetti + `games.matchWin`.
- *Catcher*: canvas 60fps, dog sprite follows finger (clamped), items spawn every 600ms (bone +1, heart +3, broccoli -2), 30s timer, combo text popups. Final score shown with `games.catcherEnd`.
- *Quiz*: 5 questions from `config.quiz`; every option allowed; correct shows green burst, wrong shows playful shake and "close enough 🤔"; always ends in a high score message.
- *Wheel*: SVG wheel, 8 segments (voucher names), drag-flick or button spin: GSAP rotation to random angle with `power4.out` over 4s, tick sfx. Result card = voucher (links to C10 collection).

C10 Vouchers. 6 scratch cards: canvas overlay (silver foil texture), pointer events erase with `destination-out` arc radius 22; when 55% cleared, auto-reveal fully with sparkle. Each voucher is a glass card with a "screenshot-ready" ratio. Stamp #7.

C11 Eggs. HUD hint "5 secrets hidden 🏠" + counter. Hit areas (invisible, 56px) added to existing elements: (1) moon in sky (draw a moon in aurora corner), (2) shooting star tap, (3) dog nose, (4) title "Riddhi" tapped 5x, (5) footer heart. Each awards once, shows `eggs[n]` toast, saves to state. Egg 2 gives a second wish input (reuse `wish.js`). Egg 5 plays `others/voice/secret.mp3` with duck. All 5 = golden paw + stamp #8.

C12 Voice. Glass card, big play/pause button, waveform: 32 bars driven by `AnalyserNode` from the audio (fallback: fake sine bars). Duck music on play, unduck on end. Playlist if several files in `config.voice`. ☒ Voice file missing = card shows "Coming soon 🎧" and Skip.

C13 Letter. Parchment-style glass card on aurora. Type lines from `config.letter` with 40ms/char, 400ms line pause, 180ms extra after commas; blinking cursor. Tap anywhere = finish instantly. Dog curled at the corner. Ends with signature fade-in (gold).

C14 Finale. Fireworks loop (every 1.2s for 10s), sky lanterns (12 rising paper-lantern divs, staggered, sway), dogs parade across bottom (`translateX` loop). Certificate card (HTML → canvas via a simple manual `<canvas>` draw so no library is needed) with "Download 🐾" button producing a PNG. Stamp row; Gold Level if complete. Final line `finale.last`. Replay menu bottom lists chapters 3 to 13.

11. config.js (all copy, owner-editable)

```
export const config = {
  name: 'Riddhi', targetISO: '2026-10-15T00:00:00+05:30', age: 22,
  web3formsKey: 'PASTE_KEY_HERE', ownerNote: 'wish -> email',
  lock: { title: 'Something special is waking up at 12:00 AM...',
    tap3: 'Shhh... wait for midnight 🌙', tap6: 'Okay okay, you\'re impatient, Miss 🥰' },
  loader: ['Loading Riddhi\'s surprise...', 'Teaching dogs to dance...', 'Hiding the cake from me..'],
  entrance: { dog: 'She\'s here!! 🐶 Happy Birthday, hooman!', button: 'Open your surprise 🎁' },
  identity: [
    {q: 'Are you Miss Alien? 🛸', a: ['Yes 🛸', 'Obviously 🛸']},
    {q: 'Are you Maalkin? 👑', a: ['Yes, bow down', 'The one and only'], runaway: 'No'},
    {q: 'Are you Maari Dobi? 🤩', a: ['Hmm... yes 🤩', 'Who told you?!'] } ],
  verified: 'VERIFIED: Riddhi Chheda, 100% real, 100% cute',
  cake: { party: 'Haha 🥳 party pooper... try again!', slice: 'Here\'s one piece for you... the real one',
  wish: { prompt: 'Close your eyes, make a wish... and write it here ✨', button: 'Send to the stars',
    note: 'Your wish will be passed to someone who will try very hard to make it come true 🐶💕',
    thanks: 'Wish received. I\'m very good at secrets... and also at making things happen 🐶' },
  balloons: [
    'Today the universe officially upgraded: Alien version 22.0 🛸', 'Warning: smiling too much to be trusted',
    'You make my 2-hour bus rides feel like 2 minutes 🚌💕', 'I\'ll never get tired of making you smile',
    'Someone believed me when it mattered. Thank you for that 🥰', 'You\'re allowed to be the main character',
    'Plot twist: you are the best part of my college life 🎓', 'You look pretty when you smile. A little',
    'Whatever you wish this year, I\'m backing you 100% 🙌', 'Surprise balloon: look in the sky, it\'s all there',
    balloonsDone: 'All popped! 🎊',
  cards: [ // front nickname, back reason
    ['Alien 🛸', 'Because you\'re one of a kind. Nobody on this planet is made like you.'],
    ['Maalkin 👑', 'Because when you say "do this", my whole world follows. 🥰'],
    ['Barbie 🧸', 'Because you light up every room, and every photo.'],
    ['Cutiee 🥰', 'Because your angry face is my favourite thing ever. Sorry, not sorry.'],
    ['Sunshine ☀️', 'Because even on my worst day, you showed up and the sky got brighter.'],
    ['Moonshine 🌙', 'Because you stay calm when everything\'s loud, and you glow in the dark.'],
    ['CR 📄', 'Because you take responsibility for everyone, even people who don\'t deserve it.'],
    ['Beauty ✨', 'Because the way you carry yourself, inside and out, is rare.'],
    ['Beautiful 🌸', 'Because I couldn\'t find a better word, so I just kept repeating it.'],
    ['Maari Dobi 🤩', 'Because only you can be this dramatic and still be adorable.'],
    ['Tap again... 🎁', 'Because you said it\'s not easy to be your best friend... and I\'m so grateful'],
  memories: [ // file, tag, caption
    ['01-same-bus', 'Same Bus 🚌', 'Where it all started. Same bus, same road, different story.'],
    ['02-kernala-fort', 'Kernala Fort Trek 🏞️', 'You struggled, I held your hand, and somehow that was the best.'],
    ['03-traditional-day', 'Traditional Day 🍛', 'I still don\'t have words for how you looked.'],
    ['04-sgnp', 'SGNP 🌿', 'Long walks, shared food, a quiet lake and cold air. Peace.'],
    ['05-ide-bootcamp', 'IDE Bootcamp, Gwalior 🏰', 'One week that made us closest.'],
    ['06-genai', 'GenAI Win 🏆', 'First of many.'],
    ['07-aavishkar', 'Aavishkar 🚀', 'From college to district level, together.'],
    ['08-synapse', 'Synapse Award 📄', 'Best Student Research Paper. Genius team.'],
    ['09-hackathon', 'Hackathon 💻', 'Sleepless, loud, unstoppable.'],
    ['10-gifts', 'Birthday Gifts 🎁', 'A perfume, a shirt, a purse, and Ganpati Bappa for the road'],
  memoriesEnd: 'There are so many more we haven\'t taken yet 💕',
  quiz: [
    {q: 'Where was our first trek?', o: ['Kernala Fort', 'Rajmachi', 'Lonavala'], c: 0},
    {q: 'What do I do that makes you angry the most?', o: ['Teasing', 'Pulling your cheeks', 'Both, obviously'], c: 0},
    {q: 'Which trip made us closest?', o: ['IDE Bootcamp, Gwalior', 'SGNP', 'Kernala'], c: 0},
    {q: 'What did I do at the trek bottom?', o: ['Held your hand', 'Ignored you', 'Ran'], c: 0},
    {q: 'Who is more dramatic?', o: ['You', 'Also you', 'Definitely you'], c: 1} ],
  vouchers: ['💎 One free treat from me. Any time, any place.', '🥰 One cute apology. Unlimited veggie hugs'],
  eggs: ['The moon winked: "She\'s the best" 🌙', 'Second wish unlocked ✨', 'Achoo! 🐶🎊', 'Riddhi\'s love is everywhere (and yes, it\'s not a joke) 🐶💕']
}
```

```
voice:[ 'others/voice/voice-note.mp3' ],
letter:['Dear Riddhi,', '', 'A year ago, you were a stranger in my mentor group.', 'Today, you ar
certificate:'Official Certificate: Riddhi Chheda is hereby declared Certified Most Special Hum
finale:{ last:'Now come get your real gift 🎁', gold:'Gold Level: Complete ⭐' }
};
```

Voice note script suggestion (about 60s, make it yours): "Happy birthday, Riddhi... I don't know how to say it properly, so I made a website. Thank you for trusting me, for the treks, for all the fights and apologies. Be happy always. I'm here."

12. Stage prompts for Codex (give one at a time)

1. **Stage 1:** "Using the spec Sections 1 to 9, create the folder structure, tokens/base CSS, state, preloader, audio engine, time lock, aurora and stars engines, and Chapters 0 and 1. Placeholder assets only. ?preview=1 bypasses the lock. Stop after, and list how to test."
2. **Stage 2:** "Add dog.js, confetti helper, HUD, Chapters 2 and 3."
3. **Stage 3:** "Add Chapter 4 (cake) fully, including mic.js with tap fallback."
4. **Stage 4:** "Add Chapters 5 and 6 including wish.js with Web3Forms and offline retry."
5. **Stage 5:** "Add Chapters 7 and 8."
6. **Stage 6:** "Add Chapters 9, 10 and 11."
7. **Stage 7:** "Add Chapters 12, 13, 14 and the certificate canvas."
8. **Stage 8 (polish):** "Audit against Section 3 rules. Fix jank, check for layout shift, check every chapter on 360x640 and 390x844 viewports, make missing assets non-fatal." After each stage ask Codex to list the acceptance tests it ran.

13. Common failure points (tell Codex to avoid)

- Autoplay: music before first tap fails on iOS. Always gate on a tap.
- 100vh jumps on mobile Safari: use 100dvh.
- backdrop-filter needs -webkit-backdrop-filter on Safari.
- Canvas blur is expensive on phones: blur the *element* with CSS, draw at half resolution.
- Memory leaks: kill timelines and listeners in `destroy()`.